

UNIVERSITY OF GHANA
DEPARTMENT STATISTICS AND ACTUARIAL SCIENCE
ASTA607 – COMPUTATIONAL STATISTICS I
CONFIDENCE ENAM SEKYERE

22427343

Question 1

```
str(Animals) library(MASS)
```

Output:

	body	brain
Mountain beaver	1.35	8.1
Cow	465.00	423.0
Grey wolf	36.33	119.5
Goat	27.66	115.0
Guinea pig	1.04	5.5
Dipliodocus	11700.00	50.0

Comment: The data is indeed a balanced data

Question 2

```
str(crabs)

par(mfrow = c(1,2)) # display side by side

hist(crabs$CL[crabs$sp=="B"],

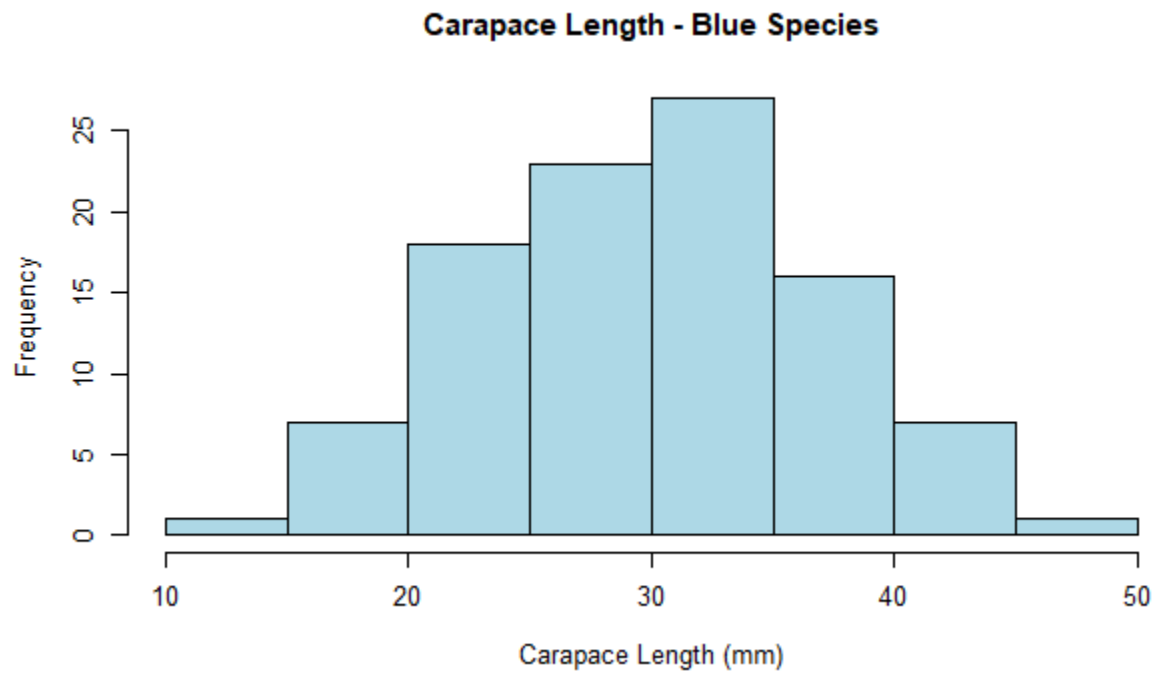
      main = "Carapace Length - Blue Species",

      xlab = "Carapace Length (mm)",

      col = "lightblue",

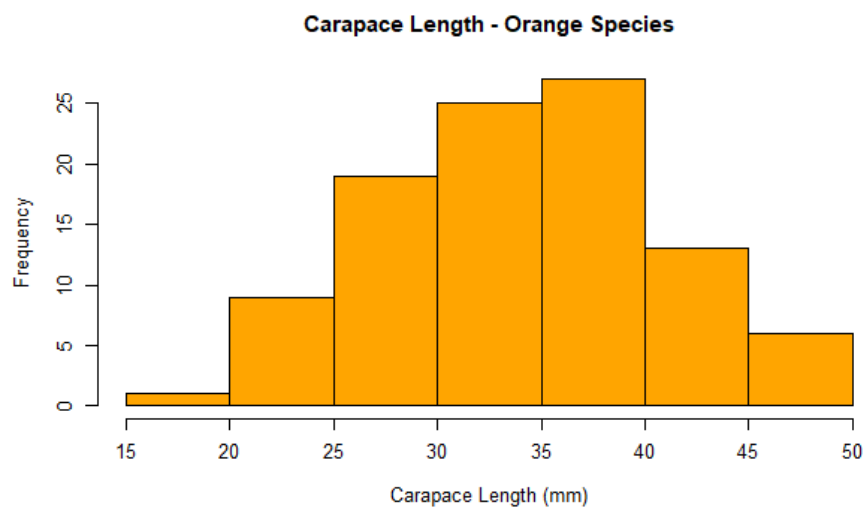
      breaks = 10)
```

```
hist(crabs$CL[crabs$sp=="B"],  
     main = "Carapace Length - Blue Species",  
     xlab = "Carapace Length (mm)",  
     col = "lightblue",  
     breaks = 10)
```



Question 3

```
hist(crabs$CL[crabs$sp=="O"],  
main = "Carapace Length - Orange Species",  
xlab = "Carapace Length (mm)",  
col = "orange",  
breaks = 10)
```



Question 4

```

barplot(t(means_matrix),

        beside = TRUE,

        legend.text = colnames(means_matrix),

        col = rainbow(5),

main = "Mean Morphological Measurements by
        Species and Sex",

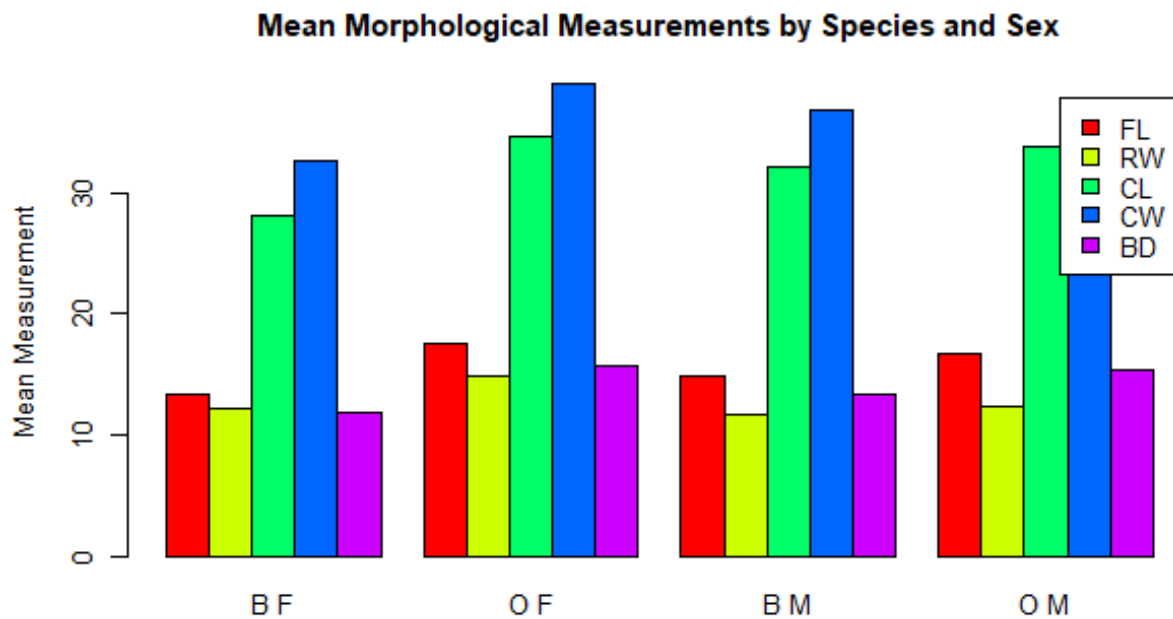
        xlab = "Species and Sex Groups",

        ylab = "Mean Measurement")

library(MASS)

data(crabs)

```



Question 5

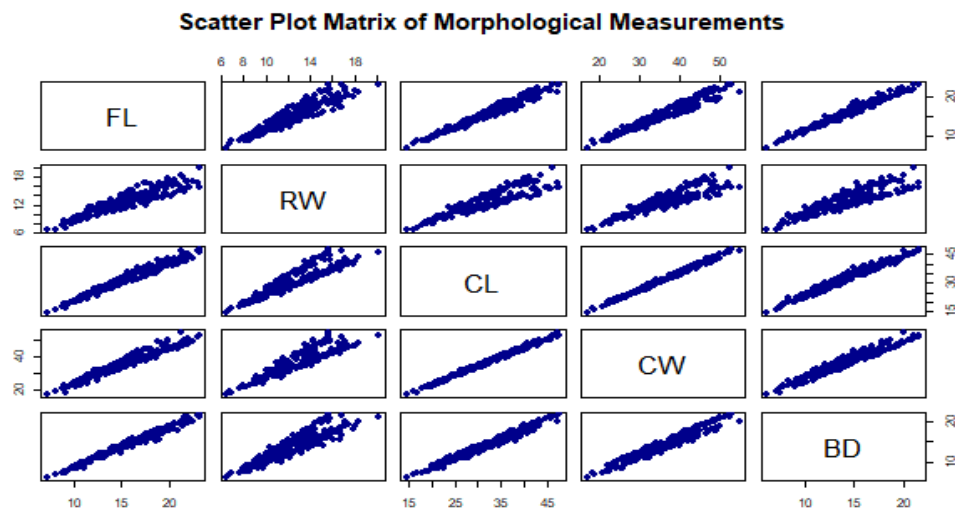
```
# Produce pairs plot

pairs(measurements,

      main = "Scatter Plot Matrix of Morphological
Measurements",

      pch = 19,

      col = "darkblue")
```



q

Strength and relationship among variables

There is an indication of very strong positive linear relationships between the variables

Question 6

Comprehensive statistical report

This report presents an analysis of morphological measurements from the **crabs** dataset in the MASS. The data consist of 200 crabs classified by **species** (Blue and Orange) and **sex** (Male and Female). For each crab, five morphological measurements were recorded

The aim of this analysis is to examine group composition, compare body size across species and sex, explore distribution of carapace length, evaluate group differences in average morphology, and assess relationships among the five measurements.